Baire Category Theorem for Complete Metric Spaces

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of baire category theorem for complete metric spaces, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

GT-04 / General Topology / Report v1

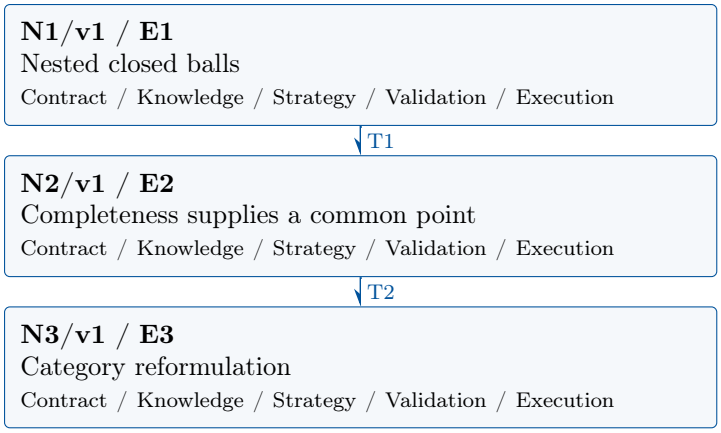
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a nonempty complete metric space X, prove every countable intersection of open dense subsets is dense. Equivalently, no nonempty open subset is a countable union of nowhere dense subsets.*

A. Foundations and permitted imports

X is a nonempty complete metric space. Dense means every nonempty open set meets the set. Nowhere dense means the closure has empty interior. Use the fact that every open neighborhood of a point contains a closed ball of sufficiently small positive radius. Compactness of such balls is not assumed.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. Nested nonempty closed balls with radii tending to 0 and all required inclusions.

Receiving activity. N2: Completeness supplies a common point

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Nested balls and shrinking radii; prior exports E1/v1

T2: N2/E2/v1 to N3. Every countable intersection of open dense sets is dense.

Receiving activity. N3: Category reformulation

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Dense-intersection theorem and nowhere-dense definition; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Nested closed balls

Object and role. Place each next ball inside the previous interior and the next dense open set.

Inputs. Open dense $G_1, G_2, \ldots$; arbitrary nonempty open U; foundational imports stated in §1

C — Contract

Own E1/v1: Nested nonempty closed balls with radii tending to 0 and all required inclusions.

K — Knowledge

Retain Open dense $G_1, G_2, \ldots$; arbitrary nonempty open U; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Place each next ball inside the previous interior and the next dense open set.

V — Validation

Do not invoke compactness of closed balls in a general complete metric space.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Choose a positive-radius closed ball B_1 contained in $U \cap G_1$, with $r_1 \leq 1/2$.

N1.2 This intersection is nonempty open.

N1.3 Inductively the open ball centered at x_n with radius r_n meets G_{n+1}; choose a point there and a positive radius $r_{n+1} \leq 2^{(-n-1)}$ so its closed ball B_{n+1} is contained in that intersection.

N1.4 Then $B_{n+1} \subset B_n$ and $B_n \subset U \cap G_1 \cap \cdots \cap G_n$.

N1.5 This works also at isolated points, where small balls may be singletons.

Export E1/v1. Nested nonempty closed balls with radii tending to 0 and all required inclusions.

Acceptance gate. Do not invoke compactness of closed balls in a general complete metric space. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Do not invoke compactness of closed balls in a general complete metric space. Recheck dependent nodes N2, N3.

B. N2/v1: Completeness supplies a common point

Object and role. Control centers by nesting and use closedness.

Inputs. Nested balls and shrinking radii; prior exports E1/v1

C — Contract

Own E2/v1: Every countable intersection of open dense sets is dense.

K — Knowledge

Retain Nested balls and shrinking radii; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Control centers by nesting and use closedness.

V — Validation

Vanishing consecutive increments alone would not have supplied this Cauchy estimate.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For $m > n$, x_m lies in B_n, so $d(x_m, x_n) \leq r_n \to 0$.

N2.2 Thus the centers are Cauchy and converge to $x \in X$ by completeness.

N2.3 For fixedn, all subsequent centers lie in the closed set B_n, so $x \in B_n$.

N2.4 Consequentlyx lies in U and every G_n.

N2.5 Since U was any nonempty open set, $\cap G_n$ is dense.

Export E2/v1. Every countable intersection of open dense sets is dense.

Acceptance gate. Vanishing consecutive increments alone would not have supplied this Cauchy estimate. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Vanishing consecutive increments alone would not have supplied this Cauchy estimate. Recheck dependent nodes N3.

C. N3/v1: Category reformulation

Object and role. Take complements of closures.

Inputs. Dense-intersection theorem and nowhere-dense definition; prior exports E2/v1

C — Contract

Own E3/v1: The stated Baire theorem and category equivalence.

K — Knowledge

Retain Dense-intersection theorem and nowhere-dense definition; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Take complements of closures.

V — Validation

The relevant sets are nowhere dense, not merely empty-interior sets without a closure check.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 If A_n are nowhere dense, $G_n = X \setminus \mathrm{cl}(A_n)$ are open dense.

N3.2 Their intersection meets every nonempty open U, so U cannot be covered by $\cup A_n$.

N3.3 Conversely, if some open dense G_n have intersection missing a nonempty open U, then $U \subset \cup(X \setminus G_n)$, each complement closed with empty interior, contradicting the category property.

N3.4 Thus the formulations are equivalent.

Export E3/v1. The stated Baire theorem and category equivalence.

Acceptance gate. The relevant sets are nowhere dense, not merely empty-interior sets without a closure check. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: The relevant sets are nowhere dense, not merely empty-interior sets without a closure check. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Nested closed balls

Fix a nonempty open set U and a sequence of open dense sets G_n.

1. Choose a positive-radius closed ball B_1 contained in $U \cap G_1$, with $r_1 \leq 1/2$.
2. This intersection is nonempty open.
3. Inductively the open ball centered at x_n with radius r_n meets G_{n+1}; choose a point there and a positive radius $r_{n+1} \leq 2^{(-n-1)}$ so its closed ball B_{n+1} is contained in that intersection.
4. Then $B_{n+1} \subset B_n$ and $B_n \subset U \cap G_1 \cap \cdots \cap G_n$.
5. This works also at isolated points, where small balls may be singletons.

B. §4.N2 — Completeness supplies a common point

1. For $m > n$, x_m lies in B_n, so $d(x_m, x_n) \leq r_n \to 0$.
2. Thus the centers are Cauchy and converge to $x \in X$ by completeness.
3. For fixedn, all subsequent centers lie in the closed set B_n, so $x \in B_n$.
4. Consequentlyx lies in U and every G_n.
5. Since U was any nonempty open set, $\cap G_n$ is dense.

C. §4.N3 — Category reformulation

1. If A_n are nowhere dense, $G_n = X \setminus \mathrm{cl}(A_n)$ are open dense.
2. Their intersection meets every nonempty open U, so U cannot be covered by $\cup A_n$.
3. Conversely, if some open dense G_n have intersection missing a nonempty open U, then $U \subset \cup(X \setminus G_n)$, each complement closed with empty interior, contradicting the category property.
4. Thus the formulations are equivalent.

□

V. WORKED ORCHESTRATION

In R, enumerate rationals q_n and set $G_n = R \setminus \{q_n\}$. Their intersection is the irrationals, which the theorem shows are dense. Each G_n is open dense; the argument constructs an irrational inside any chosen nonempty interval.

A. Boundary and hypothesis audit

Q with its usual metric is incomplete and is the countable union of singleton nowhere-dense subsets relative to Q. The complements of its enumerated points have empty total intersection. Validation traces the failure to the missing metric limit in X, not to the finite nesting steps.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

Q with its usual metric is incomplete and is the countable union of singleton nowhere-dense subsets relative to Q. The complements of its enumerated points have empty total intersection. Validation traces the failure to the missing metric limit in X, not to the finite nesting steps.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- MIT Introduction to Topology: separation, Urysohn, and Tychonoff material.